NalaWatch

Earlier, explainable decisions for urban flood preparedness.

- AI-first startup prototype | Hackathon submission
- A city operations concept connecting rainfall, drainage, exposure, and response planning.

The problem

Flood response information is fragmented.

- Heavy rainfall can overwhelm drains, flood roads, disrupt livelihoods, and create sanitation risks.
- City teams need weather, drainage, local exposure, and response plans in a shared operating picture.
- The event problem brief estimates ~US$4B annual India losses, rising to US$5B by 2030, and up to INR 400 crore for a severe disruption day in a major city. These are brief estimates, not NalaWatch impact results.

The idea

Turn separate signals into an explainable preparedness workflow.

- Combine rainfall forecasts with mapped drains, historical waterlogging, and field observations.
- Prioritize wards and corridors with visible inputs, data freshness, uncertainty, and suggested checks.
- Keep a human response owner in control of public alerts and dispatch decisions.

What works today

A small, honest, interactive prototype.

- Adjust rainfall, assumed drain load, and estimated blockage in a browser-based scenario simulator.
- See a deterministic weighted score and a matching preparedness checklist.
- Inputs and weights are illustrative. There are no live sensors, live forecast feed, trained model, or validated accuracy results.

How a pilot could work

Start narrow; validate before operational use.

- Choose one consenting city partner and a limited set of wards.
- Use permitted forecasts, drainage maps, historical incidents, and field-verified readings.
- Run in shadow mode; measure lead time, precision/recall, missed events, false alerts, and data freshness.
- Only agree on operational alerts after evaluation, human review, and incident ownership are established.

Market hypothesis

Municipal resilience and emergency operations.

- Potential users: municipal corporations, disaster management authorities, traffic cells, and infrastructure operators.
- Potential model after validation: city subscription plus integration and data services.
- Pricing, procurement path, and willingness to pay are not yet validated.

AI-first execution

AI accelerated framing, implementation, tests, and pitch materials.

- AI assistant in VS Code helped with product framing, React implementation, styling, tests, and documentation.
- Vitest, ESLint, Vite build, GitHub Actions/Linux CI, and optional Docker provide repeatable checks.
- The current risk calculation is rules-based; the deployed prototype does not call an AI model.
- Team should reconcile actual tool/model usage logs and attest to the event's 80%+ requirement.

The ask

Help us validate a safe, useful city pilot.

- Incubation support and urban hydrology expertise.
- A city or research partner with permitted historical and drainage data.
- A supervised pilot, evaluated in shadow mode before any emergency use.
- NalaWatch: prepare earlier, explain the evidence, keep people in control.